

Branes and DAA representations

Based on work with J. Huang, Y. Zhang, S. Zhuang [[arXiv:2412.19647](#)]
with S. Gukov, P. Koroteev, D. Pei, I. Saberi [[arXiv:2206.03565](#)]

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Quantization of Symplectic Manifolds

A symplectic manifold (M, ω) :

- Deformation quantization $\rightsquigarrow$ a quantum-deformed algebra $\mathcal{A}$ of functions

$$\mathcal{O}(M) \rightsquigarrow \mathcal{O}^{\hbar}(M) , \quad [\widehat{f}, \widehat{g}] = -i\hbar \widehat{\{f, g\}}.$$

- Geometric quantization $\rightsquigarrow$ a quantum Hilbert space $\mathcal{H} \cong H_{\bar{\partial}}^{0,*}(M, \mathcal{L} \otimes \sqrt{K_M})$

$$\mathfrak{B}_{\text{cc}} : \quad \begin{array}{c} \mathcal{L} \\ \downarrow \\ M \end{array} \quad c_1(\mathcal{L}) = \frac{F}{2\pi} = \frac{\omega}{2\pi\hbar}.$$

- Can we construct representation theory $\mathcal{A} \curvearrowright \mathcal{H}$ by geometry?

Brane quantizations

- Use “complexification” $\mathfrak{X}$ of the symplectic manifold M (e.g. $\mathfrak{X} = T^*M$)
- “complexification” $\mathfrak{X}$ is usually an affine hyper-Kähler manifold
- Consider A -model $\Sigma \hookrightarrow (\mathfrak{X}, \omega_{\mathfrak{X}})$
- The distinguished “big” A -brane called “canonical coisotropic brane” $\mathfrak{B}_{cc}$

$$\mathfrak{B}_{cc} : \begin{array}{c} \mathcal{L} \\ \downarrow \\ \mathfrak{X} \end{array} \quad c_1(\mathcal{L}) = [F/2\pi] \in H^2(\mathfrak{X}, \mathbb{Z}) .$$

where

$$\frac{\Omega_J}{2\pi i \hbar} = F + B + i\omega_{\mathfrak{X}}$$

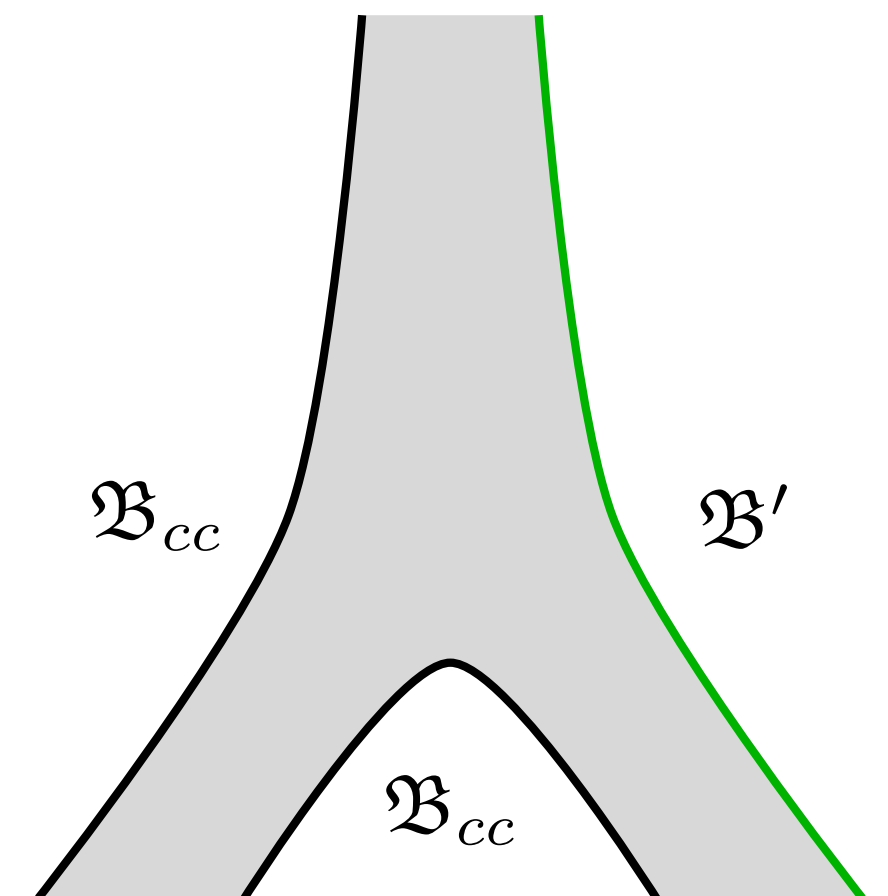
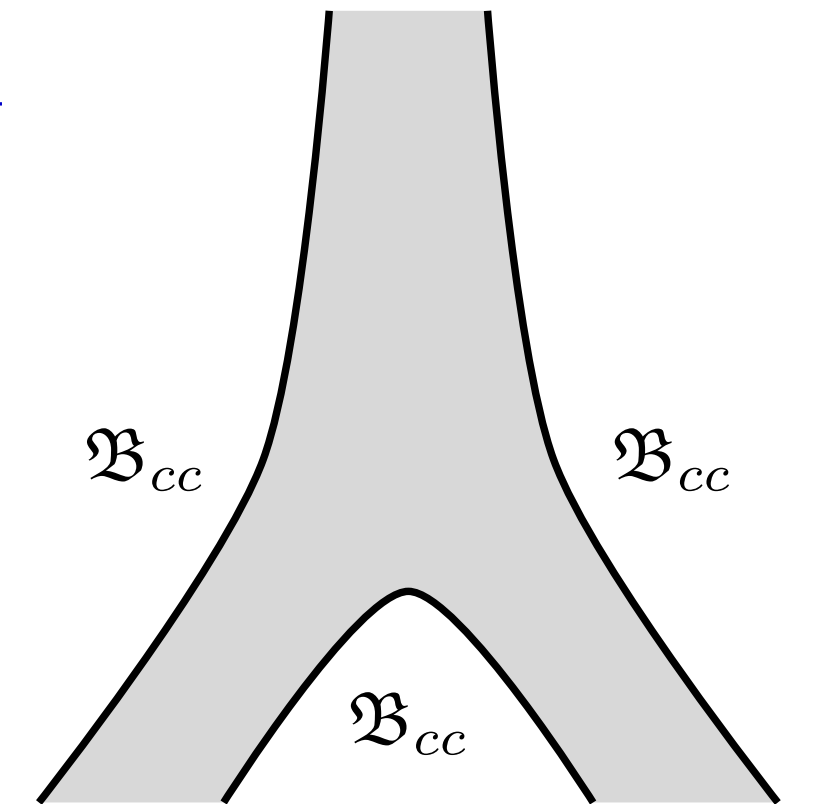
- M is a Lagrangian with respect to $\omega_{\mathfrak{X}}$, so can consider an ordinary A -brane on M

$$\mathfrak{B}_M : \begin{array}{c} \mathcal{L}' \\ \downarrow \\ M \end{array} \quad \text{local system}$$

Brane quantizations

- **Brane quantization** bridges **deformation quantization** and **geometric quantization**
- $(\mathfrak{B}_{cc}, \mathfrak{B}_{cc})$ -string forms an algebra $\mathcal{A}$ by joining strings
- $\text{Hom}(\mathfrak{B}_{cc}, \mathfrak{B}_{cc})$ provides **deformation quantization** of $\mathcal{O}(\mathfrak{X})$ with respect to Ω_J
- $(\mathfrak{B}_{cc}, \mathfrak{B}_M)$ -string receives an action of $(\mathfrak{B}_{cc}, \mathfrak{B}_{cc})$ -string
- $\text{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc})$ provides **geometric quantization** of M
- This picture provides **representation theory** of $\mathcal{A}$ via **A-model** on $\mathfrak{X}$:

$$\begin{aligned} \mathcal{A} &= \text{Hom}(\mathfrak{B}_{cc}, \mathfrak{B}_{cc}) \\ \mathcal{H} &= \text{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc}) . \end{aligned}$$



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compact A-branes $\longleftrightarrow$ finite-dimensional $\mathcal{A}$ -modules
 non-compact A-branes $\longleftrightarrow$ infinite-dimensional $\mathcal{A}$ -modules

Derived functor

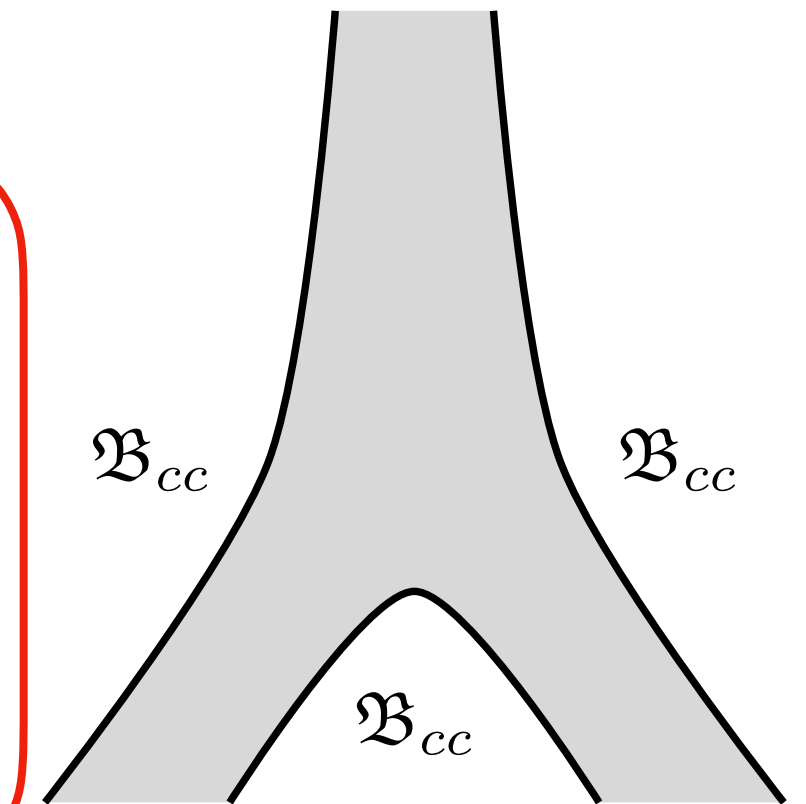
- Brane quantization provides a **derived functor**

Physics

$$\mathbf{R}\mathrm{Hom}(-, \mathfrak{B}_{cc}) : D^b A\text{-Brane}(\mathfrak{X}, \omega_{\mathfrak{X}}) \rightarrow D^b \mathrm{Rep}(\mathcal{O}^q(\mathfrak{X}))$$

Geometric category

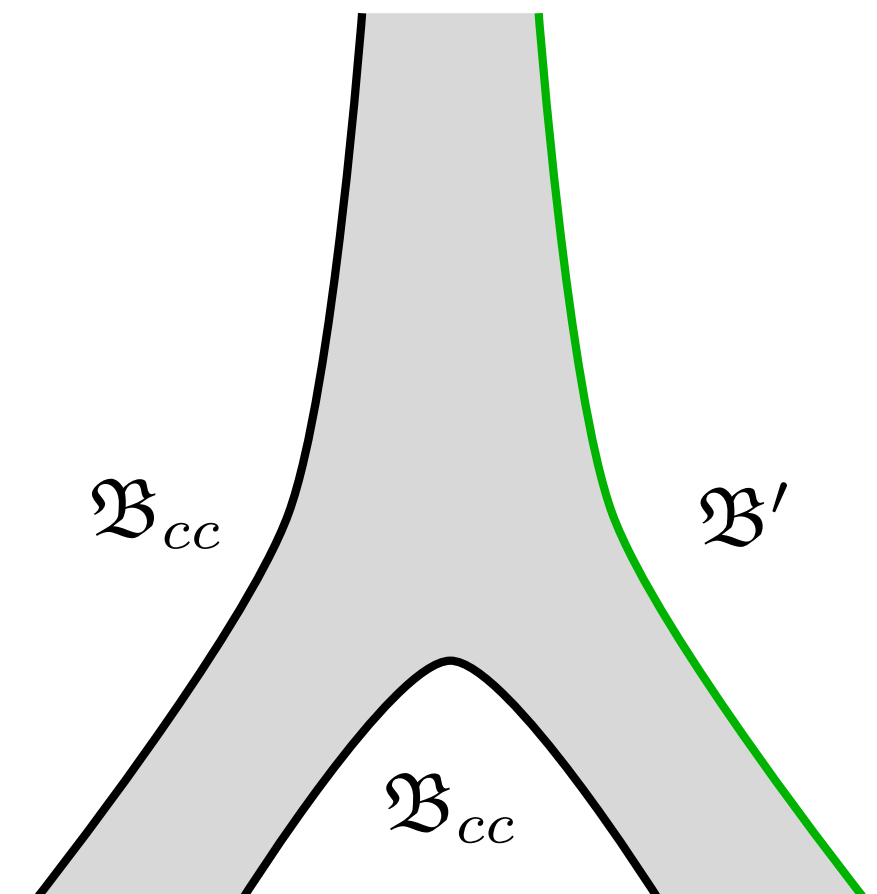
Representation category



- This picture provides **representation theory** of $\mathcal{A}$ via **A-model** on $\mathfrak{X}$:

$$\mathcal{A} = \mathrm{Hom}(\mathfrak{B}_{cc}, \mathfrak{B}_{cc})$$

$$\mathcal{H} = \mathrm{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc}) .$$



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compact A -branes $\longleftrightarrow$ finite-dimensional $\mathcal{A}$ -modules

non-compact A -branes $\longleftrightarrow$ infinite-dimensional $\mathcal{A}$ -modules

Our target space

- Moduli space of flat $\text{SL}(2, \mathbb{C})$ -connections on a four-punctured sphere $C_{0,4}$

$$\mathfrak{X} = \mathcal{M}_{\text{flat}}(C_{0,4}, \text{SL}(2, \mathbb{C})) = \langle M_1, M_2, M_3, M_4 \in \text{SL}(2, \mathbb{C}) \mid M_1 M_2 M_3 M_4 = \text{Id} \rangle / \text{SL}(2, \mathbb{C}) ,$$

- These variables are subject to the **cubic equation**

$$f(x, y, z) = -xyz + x^2 + y^2 + z^2 + \theta_1 x + \theta_2 y + \theta_3 z + \theta_4 = 0 ,$$

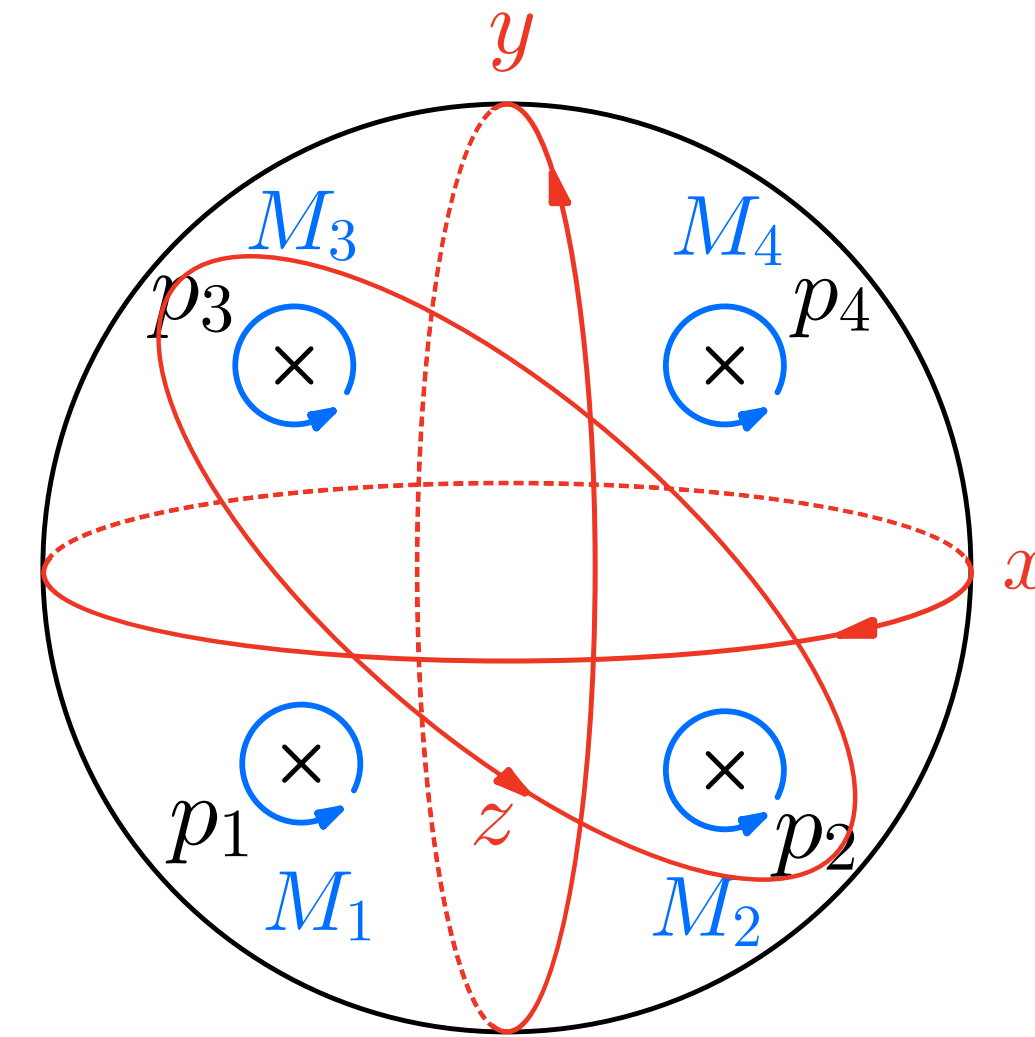
where, using the notation $\bar{f} \equiv f + f^{-1}$, θ_i are expressed by

$$\theta_1 = \chi_{D_4}(\mathbf{8}_V) = \bar{t}_1 \bar{t}_2 + \bar{t}_3 \bar{t}_4 ,$$

$$\theta_2 = \chi_{D_4}(\mathbf{8}_S) = \bar{t}_1 \bar{t}_3 + \bar{t}_2 \bar{t}_4 ,$$

$$\theta_3 = \chi_{D_4}(\mathbf{8}_C) = \bar{t}_1 \bar{t}_4 + \bar{t}_2 \bar{t}_3 ,$$

$$\theta_4 = \chi_{D_4}(\mathbf{adj}) = \bar{t}_1^2 + \bar{t}_2^2 + \bar{t}_3^2 + \bar{t}_4^2 + \bar{t}_1 \bar{t}_2 \bar{t}_3 \bar{t}_4 - 4 .$$



- This space is the Coulomb branch of 4d $\mathcal{N} = 2$ $\text{SU}(2)$ $N_f = 4$ Seiberg-Witten theory on $S^1 \times \mathbb{R}^3$
- Flavor symmetry is $\text{SO}(8)$

Hitchin moduli space

- Moduli space of parabolic $SU(2)$ -Higgs bundles on a four-punctured sphere $C_{0,4}$

$$\mathcal{M}_H(C_{0,4}, SU(2)) = \{(E, \varphi) \mid E \text{ holom. bdle}, \varphi \in H^0(C_{0,4}, K^{\otimes 2})\}_{\text{stab. cond.}},$$

- Tame ramifications $(\alpha_i, \beta_i, \gamma_i) \in (\mathfrak{t}_{A_1})^3$ at punctures $i = (1, 2, 3, 4)$

$$A = \alpha_j d\vartheta + \dots$$

$$\varphi = \frac{1}{2}(\beta_j + i\gamma_j) \frac{dz}{z} + \dots$$

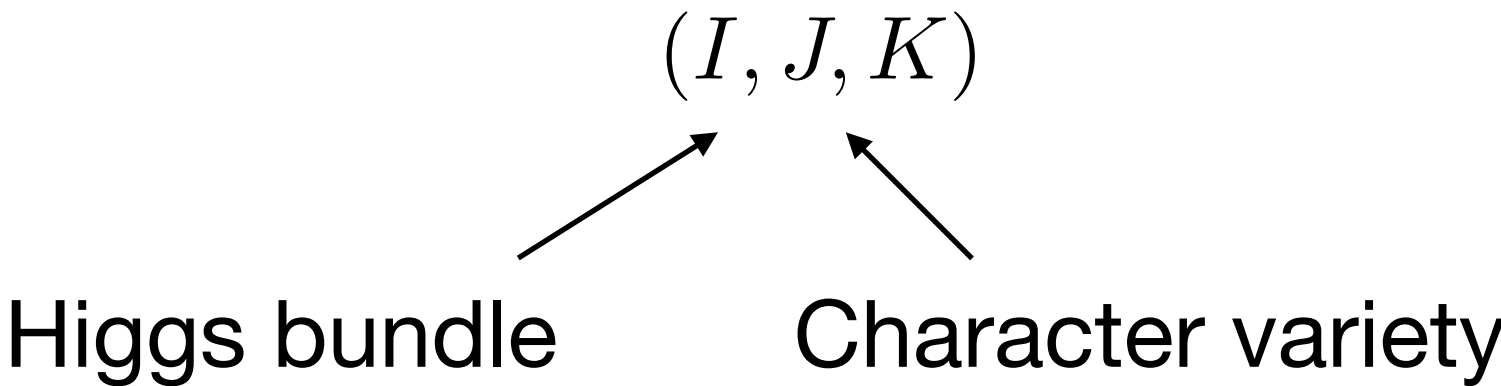
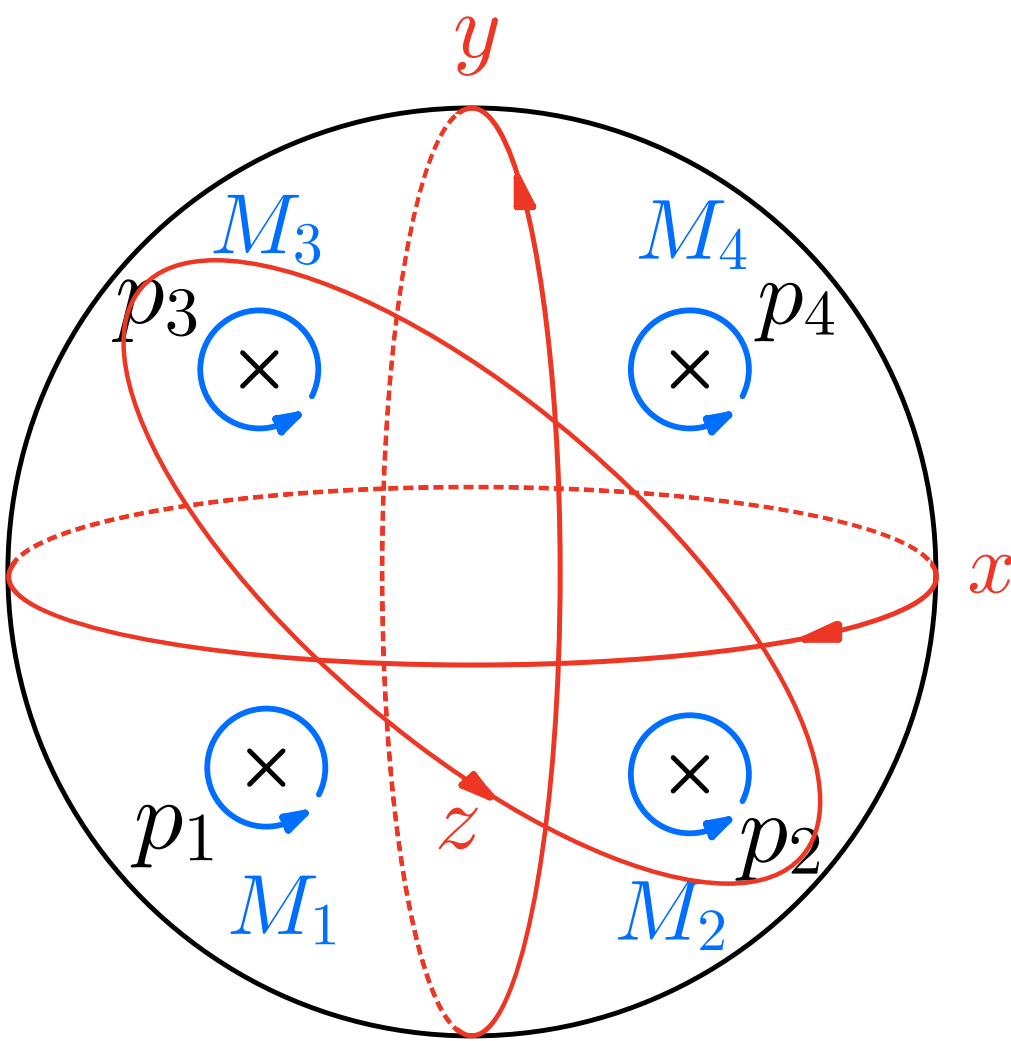
- Ramification data are enhanced to the Cartan subalgebra of D_4

$$\prod_{j=1}^4 \mathfrak{t}_j^{(\alpha)} \times \mathfrak{t}_j^{(\beta)} \times \mathfrak{t}_j^{(\gamma)} \cong \mathfrak{t}_{D_4}^{(\alpha)} \times \mathfrak{t}_{D_4}^{(\beta)} \times \mathfrak{t}_{D_4}^{(\gamma)}$$

- Non-Abelian Hodge correspondence: $\mathfrak{X} = \mathcal{M}_{\text{flat}}(C_{0,4}, SL(2, \mathbb{C})) \cong \mathcal{M}_H(C_{0,4}, SU(2))$

- Complex structures as a hyper-Kähler manifold

$$t_j = \exp(-2\pi(\gamma_j + i\alpha_j)), \quad (j = 1, 2, 3, 4)$$



Double Affine Hecke Algebra

DAHA of $C^\vee C_1$

Cherednik, Noumi, van Diejen, Sahi, Stokman

$$\begin{aligned}(T_0 - t_1) (T_0 + t_1^{-1}) &= 0, \\ (T_0^\vee - t_2) (T_0^\vee + t_2^{-1}) &= 0, \\ (T_1 - t_3) (T_1 + t_3^{-1}) &= 0, \\ (T_1^\vee - t_4) (T_1^\vee + t_4^{-1}) &= 0,\end{aligned}$$

and

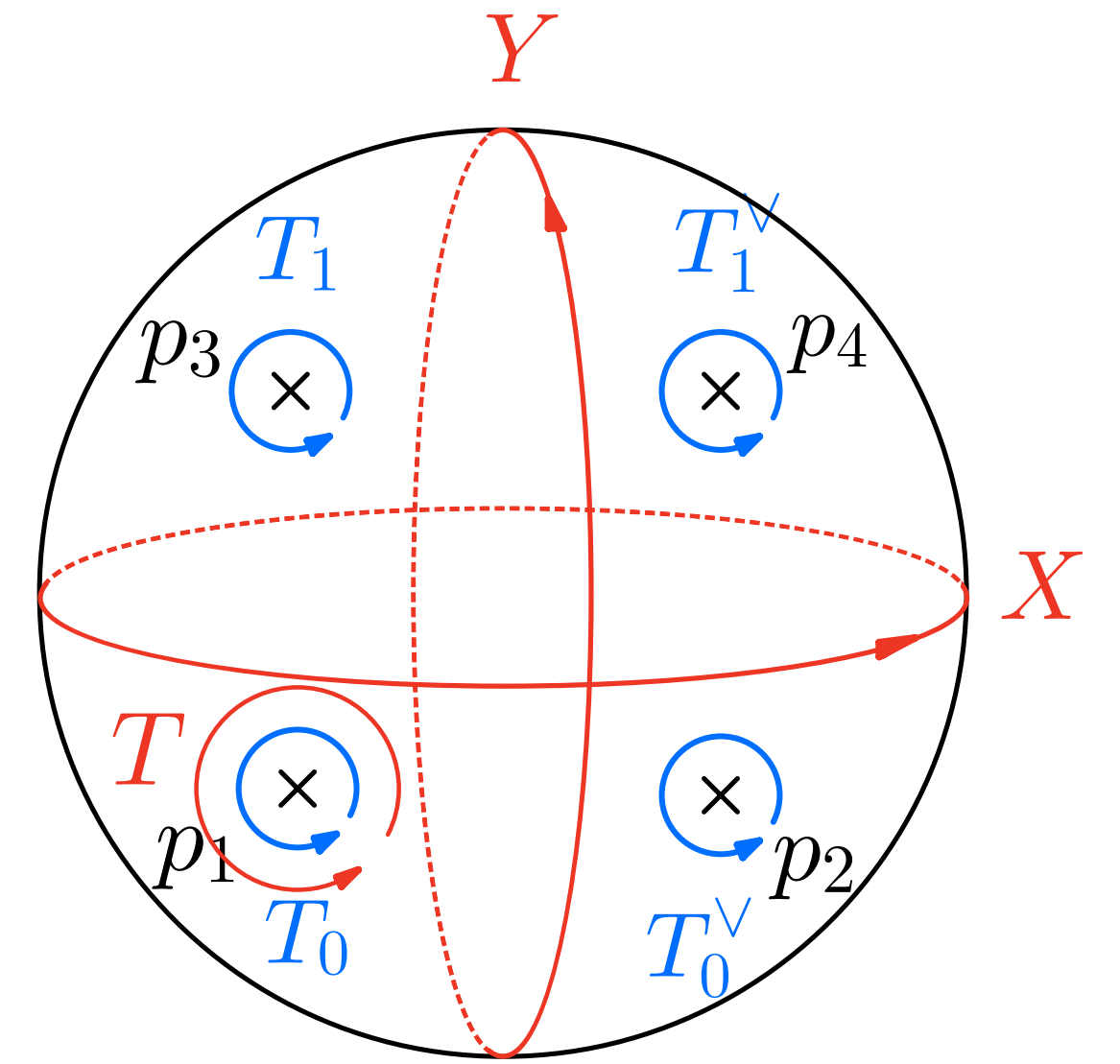
$$T_1^\vee T_1 T_0 T_0^\vee = q^{-\frac{1}{2}}.$$

The **spherical DAHA** is then defined by the idempotent projection

$$S\ddot{H}_{q,t} = \mathbf{e}\ddot{H}_{q,t}\mathbf{e} , \quad \mathbf{e} = \frac{T_0 + t_1^{-1}}{t_1 + t_1^{-1}} .$$

Generators

$$\begin{aligned}x &= (T_0^\vee T_0 + (T_0^\vee T_0)^{-1}) \mathbf{e} = (X + X^{-1}) \mathbf{e} , \\ y &= (T_0 T_1 + (T_0 T_1)^{-1}) \mathbf{e} = (Y + Y^{-1}) \mathbf{e} , \\ z &= (T_0 T_1^\vee + (T_0 T_1^\vee)^{-1}) \mathbf{e} .\end{aligned}$$



Spherical DAHA

The q -deformed algebra relations are given by:

$$\begin{aligned} [x, y]_q &= (q - q^{-1})z + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_3 , \\ [y, z]_q &= (q - q^{-1})x + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_1 , \\ [z, x]_q &= (q - q^{-1})y + (q^{\frac{1}{2}} - q^{-\frac{1}{2}})\theta_2 , \\ -q^{\frac{1}{2}}xyz + qx^2 + q^{-1}y^2 + qz^2 + q^{\frac{1}{2}}\theta_1x + q^{-\frac{1}{2}}\theta_2y + q^{\frac{1}{2}}\theta_3z + \theta_4(q) &= 0 , \end{aligned}$$

where, using the notation $\bar{f} \equiv f + f^{-1}$, θ_i are expressed by

$$\begin{aligned} \theta_1 &= \chi_{D_4}(\mathbf{8}_V) = \bar{t}_1 \bar{t}_2 + \bar{t}_3 \bar{t}_4 , \\ \theta_2 &= \chi_{D_4}(\mathbf{8}_S) = \bar{t}_1 \bar{t}_3 + \bar{t}_2 \bar{t}_4 , \\ \theta_3 &= \chi_{D_4}(\mathbf{8}_C) = \bar{t}_1 \bar{t}_4 + \bar{t}_2 \bar{t}_3 , \\ \theta_4(q) &= \bar{t}_1^2 + \bar{t}_2^2 + \bar{t}_3^2 + \bar{t}_4^2 + \bar{t}_1 \bar{t}_2 \bar{t}_3 \bar{t}_4 - q - q^{-1} - 2 . \end{aligned}$$

The A_1 limit is $\theta_{1,2,3} = 0$.

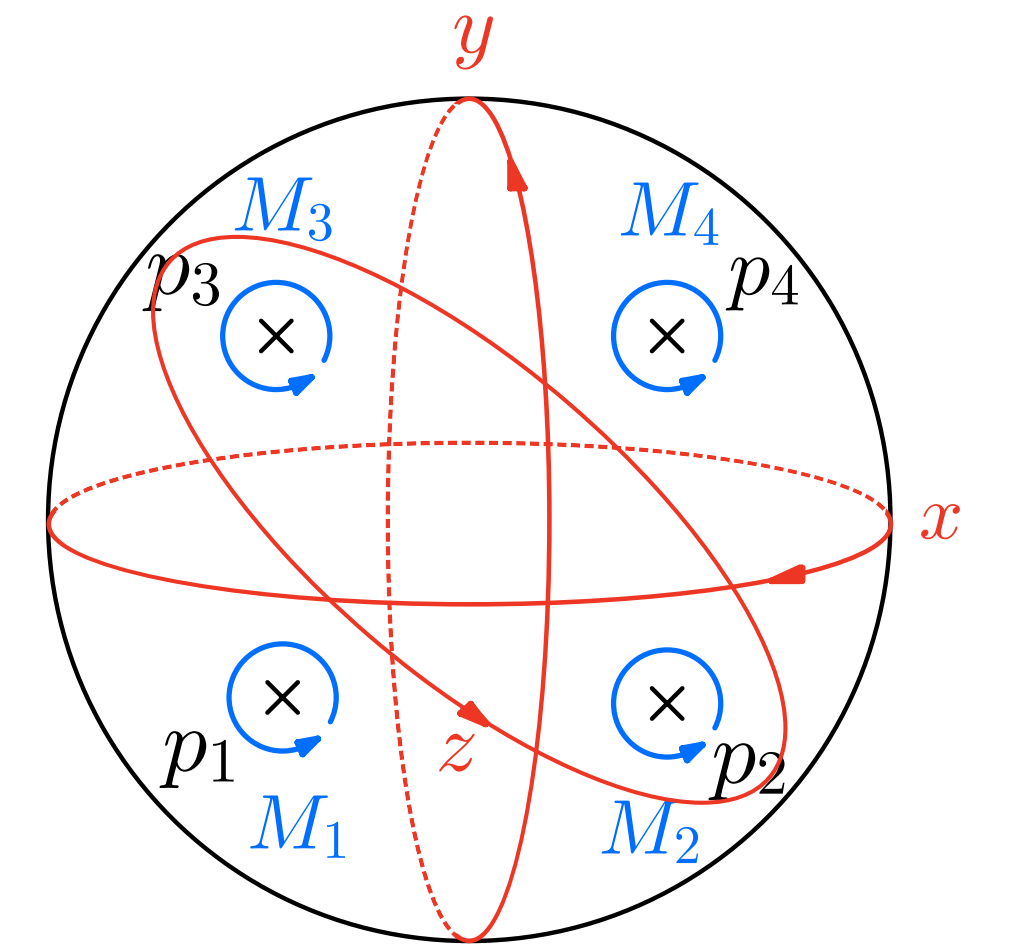
Symmetry

- Weyl group $W(D_4)$
- Braid group B_3
- Sign-flip group $\mathbb{Z}_2^{\times 2}$



$S\ddot{H}_{q,t}$ is the deformation quantization of $\mathcal{O}(\mathfrak{X})$ with respect to Ω_J !

$$\begin{aligned} \tau_+ : (x, y, z, \theta_1, \theta_2, \theta_3, \theta_4) &\mapsto \left(x, q^{-\frac{1}{2}}(xy - q^{-\frac{1}{2}}z - \theta_3), y, \theta_1, \theta_3, \theta_2, \theta_4 \right) , \\ \tau_- : (x, y, z, \theta_1, \theta_2, \theta_3, \theta_4) &\mapsto \left(q^{-\frac{1}{2}}(xy - q^{-\frac{1}{2}}z - \theta_3), y, x, \theta_3, \theta_2, \theta_1, \theta_4 \right) , \\ \sigma : (x, y, z, \theta_1, \theta_2, \theta_3, \theta_4) &\mapsto \left(y, x, q^{-\frac{1}{2}}(xy - q^{-\frac{1}{2}}z - \theta_3), \theta_2, \theta_1, \theta_3, \theta_4 \right) . \end{aligned}$$



Physics

- The **coordinate ring** of the **Coulomb branch** is an algebra of **line operators** of **4d** $\mathcal{N} = 2$ theory

$x \longleftrightarrow$ Wilson loop ,

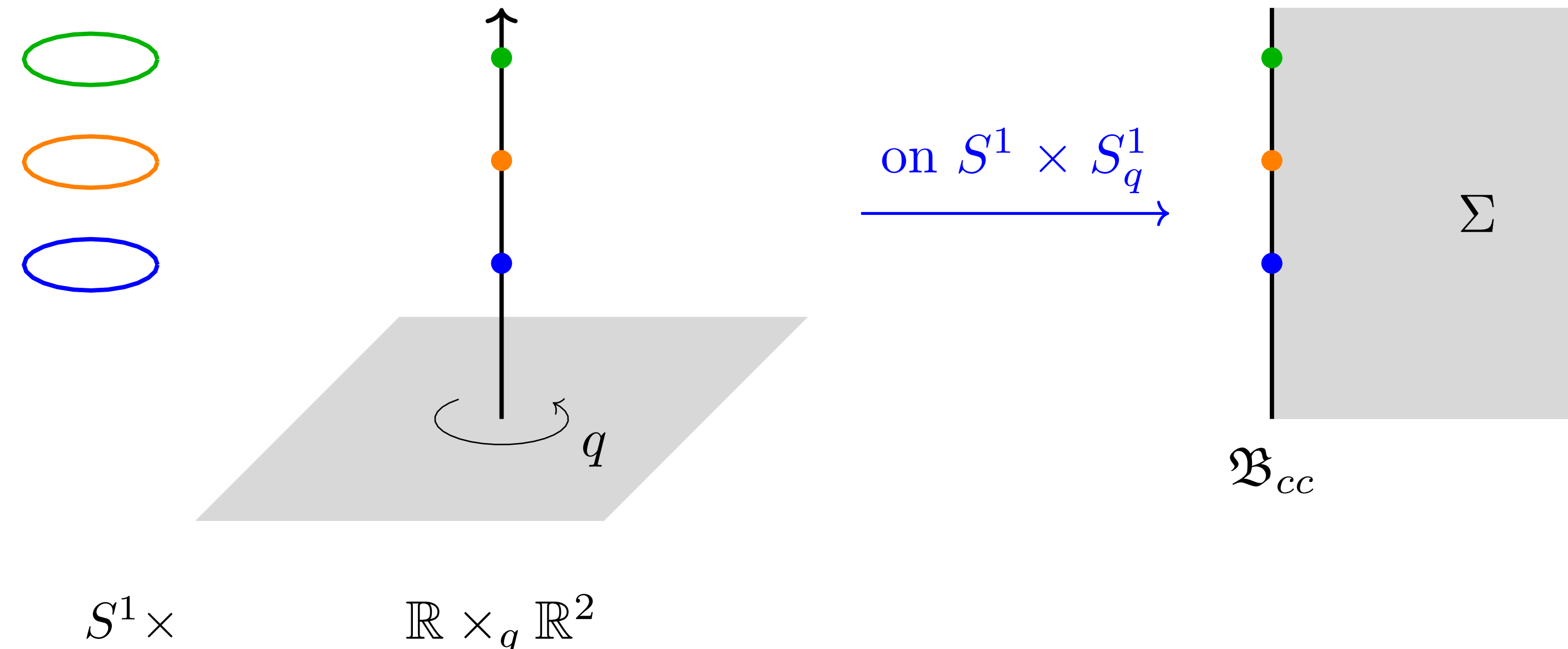
$y \longleftrightarrow$ 't Hooft loop ,

$z \longleftrightarrow$ dyonic loop .

- Ω -deformation introduces non-commutativity
- $S\ddot{H}_{q,t}$ is the algebra of **line operators** in $SU(2)$ $N_f = 4$ theory on $S^1 \times \mathbb{R} \times_q \mathbb{R}^2$
- Compactification from 4d to 2d

↓

$\mathfrak{B}_{cc}$ boundary condition



Our Goal

Our goal is to present evidence of derived equivalence

$$\mathbf{R}\mathrm{Hom}(-, \mathfrak{B}_{cc}) : D^b A\text{-Brane}(\mathfrak{X}, \omega_{\mathfrak{X}}) \rightarrow D^b \mathrm{Rep}(S\ddot{H}_{q,t}),$$
$$\mathfrak{B}_M \mapsto \mathcal{H} = \mathrm{Hom}(\mathfrak{B}_M, \mathfrak{B}_{cc}) .$$

A-brane category $\longleftrightarrow$ Representation category

Brane Quantization

Polynomial representation

The polynomial representation is given by the homomorphism

$$\begin{aligned} \text{pol} : S\ddot{H}_{q,t} &\rightarrow \text{End}(\mathcal{P}) , \\ x &\mapsto X + X^{-1} , \\ y &\mapsto A(X; \mathbf{t})(\varpi - 1) + A(X^{-1}; \mathbf{t})(\varpi^{-1} - 1) - q^{\frac{1}{2}}t_1t_3 - q^{-\frac{1}{2}}(t_1t_3)^{-1} , \\ z &\mapsto q^{\frac{1}{2}} [XA(X)\varpi + X^{-1}A(X^{-1})\varpi^{-1}] \\ &\quad - \frac{X + X^{-1}}{q^{\frac{1}{2}} + q^{-\frac{1}{2}}} \left[A(X) + A(X^{-1}) + q^{\frac{1}{2}}t_1t_3 + q^{-\frac{1}{2}}t_1^{-1}t_3^{-1} \right] - \frac{\theta_3}{q^{\frac{1}{2}} + q^{-\frac{1}{2}}} . \end{aligned}$$

where

$$A(X; \mathbf{t}) = - \frac{(1 + q^{\frac{1}{2}}t_1t_2X)(1 + q^{\frac{1}{2}}t_1X/t_2)(1 + q^{\frac{1}{2}}t_3t_4X)(1 + q^{\frac{1}{2}}t_3X/t_4)}{q^{\frac{1}{2}}t_1t_3(1 - X^2)(1 - qX^2)} ,$$

Askey-Wilson polynomial

$$P_n(X; q, \mathbf{t}) := \frac{(qab, qac, qad; q)_n}{q^{\frac{n}{2}}a^n(abcdq^{n+1}; q)_n} {}_4\phi_3 \left(\begin{matrix} q^{-n}, q^{n+1}abcd, q^{\frac{1}{2}}aX, q^{\frac{1}{2}}aX^{-1} \\ qab, qac, qad \end{matrix} \middle| q; q \right) ,$$

is the eigenfunction of y under polynomial representation

$$\text{pol}(y) \cdot P_n(x; q, \mathbf{t}) = (-q^{n+\frac{1}{2}}t_1t_3 - q^{-n-\frac{1}{2}}t_1^{-1}t_3^{-1})P_n(x; q, \mathbf{t})$$

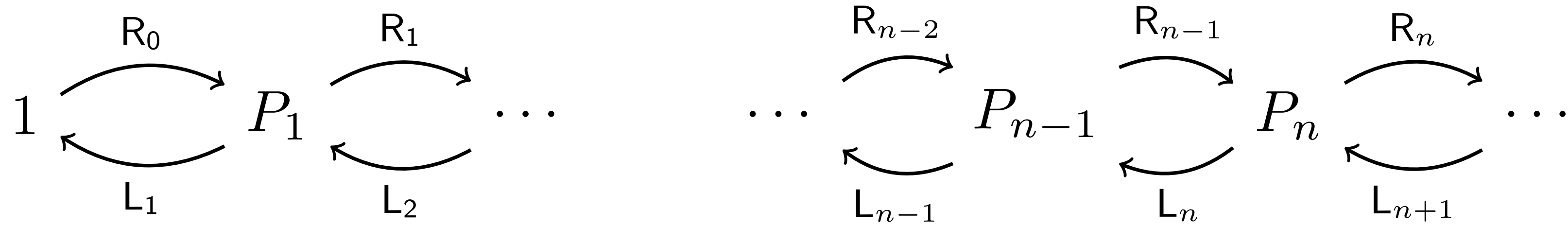
(A,B,A)-branes and polynomial representation

Claim The (A, B, A) -branes supported on 24 lines in the affine cubic surface $\mathfrak{X}$ correspond to the polynomial representation of $S\ddot{H}_{q,t}$ and its images under certain symmetries.

Eight lines on the plane where y is constant are denoted by $\mathbf{P}_w$, where $w \in \mathbf{P}(8_S)$.

$$\begin{aligned}\mathbf{P}_{(1,0,1,0)} &= \{y = -t_1 t_3 - t_1^{-1} t_3^{-1}, z = -t_1 t_3 x - t_1 t_4 - t_1 t_4^{-1} - t_2 t_3 - t_2^{-1} t_3\}, \\ \mathbf{P}_{(-1,0,-1,0)} &= \{y = -t_1 t_3 - t_1^{-1} t_3^{-1}, z = -t_1^{-1} t_3^{-1} x - t_1^{-1} t_4 - t_1^{-1} t_4^{-1} - t_2 t_3^{-1} - t_2^{-1} t_3^{-1}\}, \\ \mathbf{P}_{(-1,0,1,0)} &= \{y = -t_1^{-1} t_3 - t_1 t_3^{-1}, z = -t_1^{-1} t_3 x - t_2 t_3 - t_2^{-1} t_3 - t_1^{-1} t_4 - t_1^{-1} t_4^{-1}\}, \\ \mathbf{P}_{(1,0,-1,0)} &= \{y = -t_1^{-1} t_3 - t_1 t_3^{-1}, z = -t_1 t_3^{-1} x - t_1 t_4 - t_1 t_4^{-1} - t_2 t_3^{-1} - t_2^{-1} t_3^{-1}\}, \\ \mathbf{P}_{(0,1,0,1)} &= \{y = -t_2 t_4 - t_2^{-1} t_4^{-1}, z = -t_2 t_4 x - t_2 t_3 - t_2 t_3^{-1} - t_1 t_4 - t_1^{-1} t_4\}, \\ \mathbf{P}_{(0,-1,0,-1)} &= \{y = -t_2 t_4 - t_2^{-1} t_4^{-1}, z = -t_2^{-1} t_4^{-1} x - t_2^{-1} t_3 - t_2^{-1} t_3^{-1} - t_1 t_4^{-1} - t_1^{-1} t_4^{-1}\}, \\ \mathbf{P}_{(0,-1,0,1)} &= \{y = -t_2^{-1} t_4 - t_2 t_4^{-1}, z = -t_2^{-1} t_4 x - t_2^{-1} t_3 - t_1 t_4 - t_1^{-1} t_4 - t_2^{-1} t_3^{-1}\}, \\ \mathbf{P}_{(0,1,0,-1)} &= \{y = -t_2^{-1} t_4 - t_2 t_4^{-1}, z = -t_2 t_4^{-1} x - t_1 t_4^{-1} - t_2 t_3 - t_2 t_3^{-1} - t_1^{-1} t_4^{-1}\}.\end{aligned}$$

Raising and Lowering operators



The raising/lowering operators are given by:

$$R_n = -q^{1-n}x - qt_1t_3z + E_n ,$$

$$L_n = -q^{n+2}x - \frac{q}{t_1t_3}z + D_n ,$$

where

$$D_n = \frac{q^{3/2}t_1^{-1}t_3^{-1} (q^{-n}t_1^{-1}t_3^{-1}\theta_1 - \theta_3)}{1 - q^{-2n}t_1^{-2}t_3^{-2}} ,$$

$$E_n = \frac{q^{3/2}t_1t_3 (q^{n+1}t_1t_3\theta_1 - \theta_3)}{q^{2n+2}t_1^2t_3^2 - 1} .$$

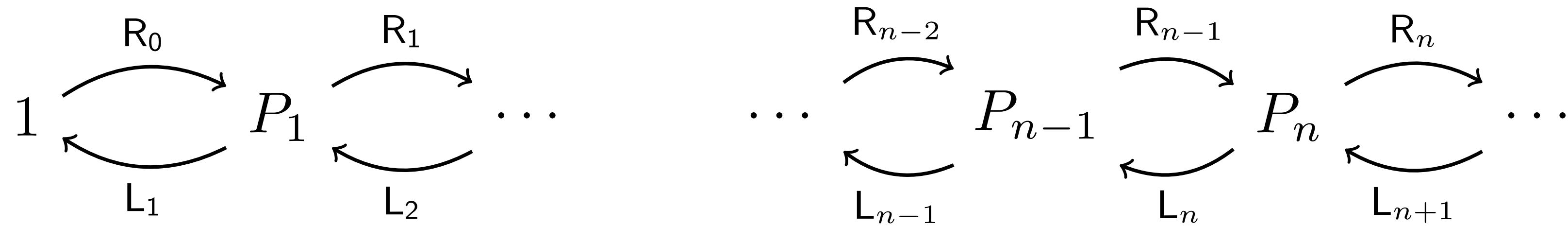
Under the polynomial representation,

$$\text{pol}(R_n) \cdot P_n(x; q, \mathbf{t}) = q^{1-n} (t_1^2 t_3^2 q^{2n+1} - 1) P_{n+1}(x; q, \mathbf{t}) ,$$

$$\text{pol}(L_n) \cdot P_n(x; q, \mathbf{t}) = -q^{1-n} (q^n - t_1^{-1} t_2 t_3^{-1} t_4) (q^n - t_1^{-1} t_2 t_3^{-1} t_4^{-1}) (q^n - t_1^{-1} t_2^{-1} t_3^{-1} t_4)$$

$$\times (q^n - t_1^{-1} t_2^{-1} t_3^{-1} t_4^{-1}) \frac{(q^n - 1)(q^n - t_1^{-2})(q^n - t_3^{-2})(q^n - t_1^{-2} t_3^{-2})}{(q^{2n} - t_1^{-2} t_3^{-2})^2 (q^{2n-1} - t_1^{-2} t_3^{-2})} P_{n-1}(x; q, \mathbf{t}) .$$

Finite-dimensional representations



- Using the **Weyl group symmetry** $W(D_4)$, all the **shortening conditions** for **finite-dimensional modules** obtained in this way can be neatly repackaged as

$$q^n = t_1^{-r_1} t_2^{-r_2} t_3^{-r_3} t_4^{-r_4} =: \mathbf{t}^{-r}, \quad r \in R(D_4) \cup \{(0, 0, 0, 0)\} .$$

- From **brane quantization**, the **dimension** of the **module** is computed

$$\begin{aligned} \dim \mathcal{H} &= \dim H^0(M, \mathfrak{B}_{\text{cc}} \otimes \mathfrak{B}_M^{-1}) \\ &= \int_M \text{ch}(\mathfrak{B}_{\text{cc}}) = \int_M \frac{F + B}{2\pi} , \end{aligned}$$

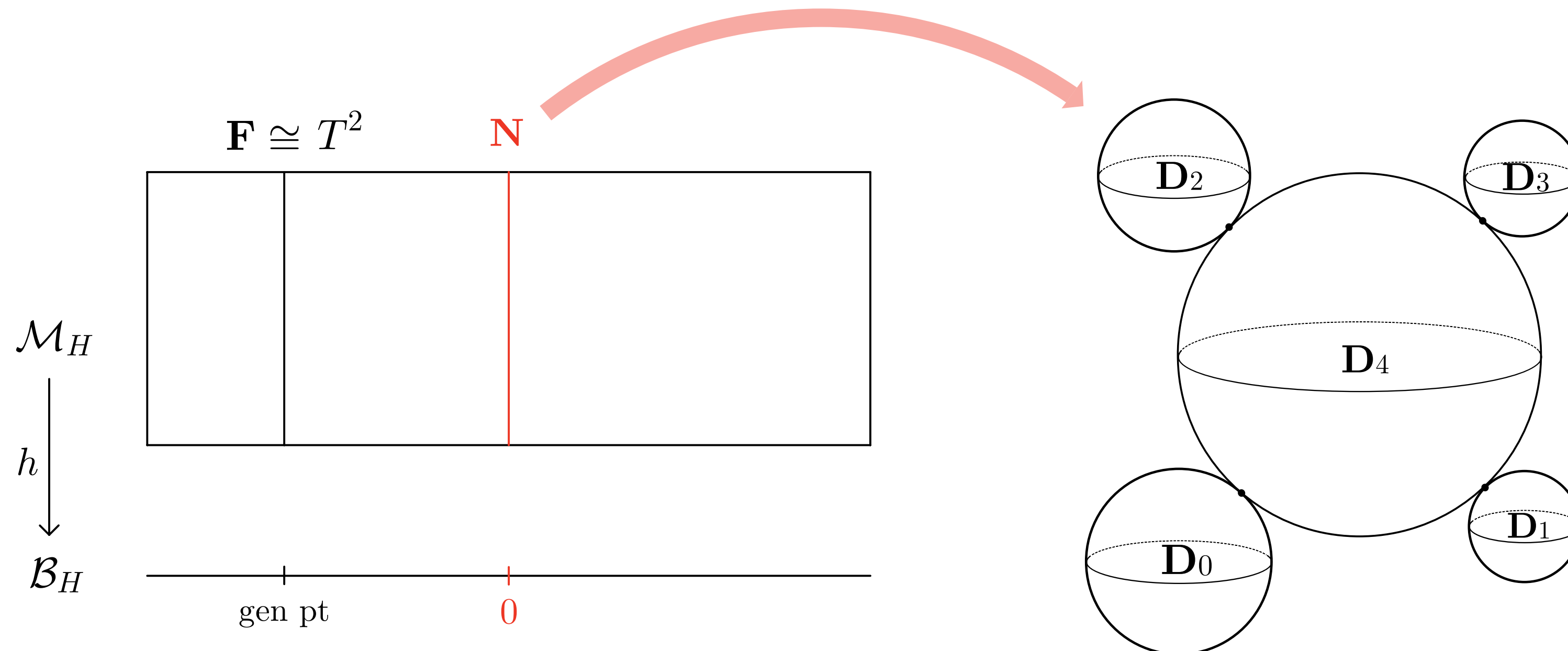
Hitchin fibration

- In complex structure I , the Hitchin moduli space admits the Hitchin fibration over $\mathcal{B}_H$:

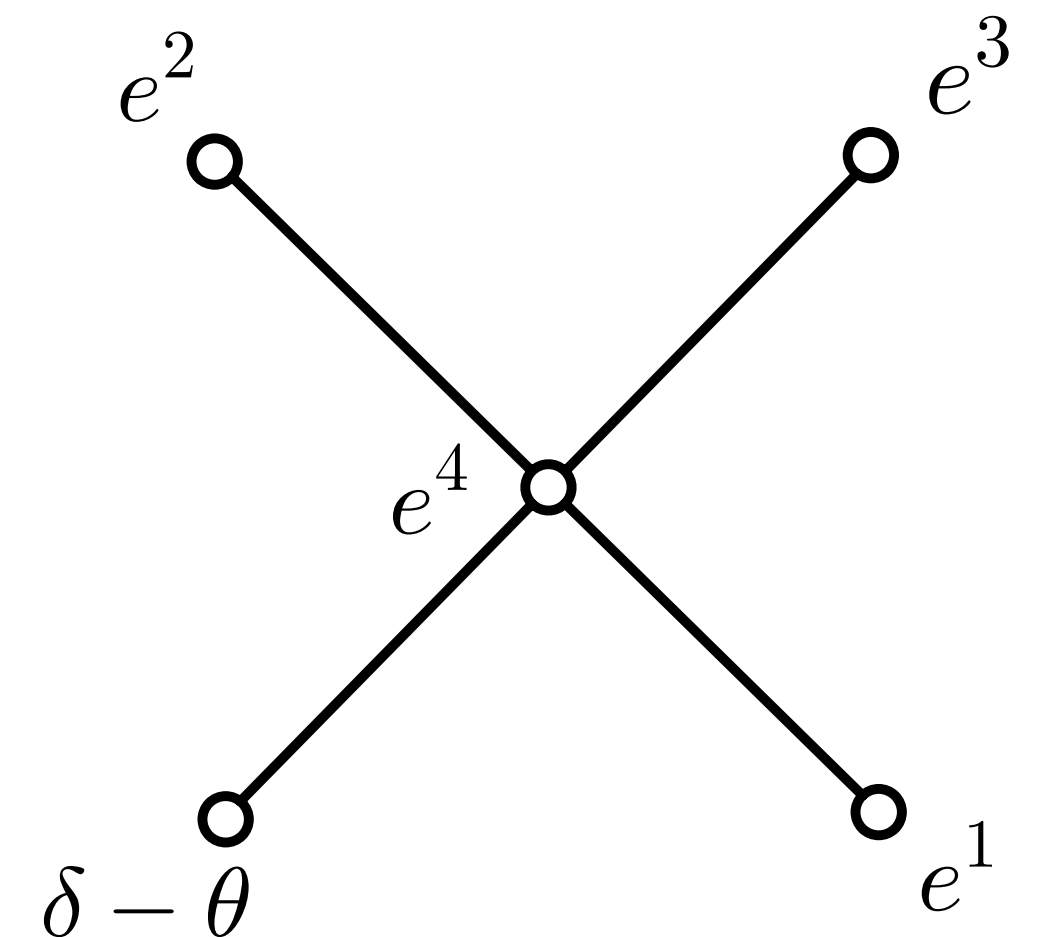
$$h : \mathcal{M}_H(C_{0,4}, \mathrm{SU}(2)) \rightarrow \mathcal{B}_H = H^0(C_{0,4}, K_C^{\otimes 2}) .$$

$$(E, \varphi) \mapsto \mathrm{Tr}(\varphi^2)$$

- In our case, the Hitchin fibration is an elliptic fibration with a singular fiber.
- When $\beta_i = 0 = \gamma_i$, the singular fiber is located at the origin.
- fiber class relation: $[\mathbf{F}] = 2[\mathbf{D}_4] + \sum_{i=0}^3 [\mathbf{D}_i]$



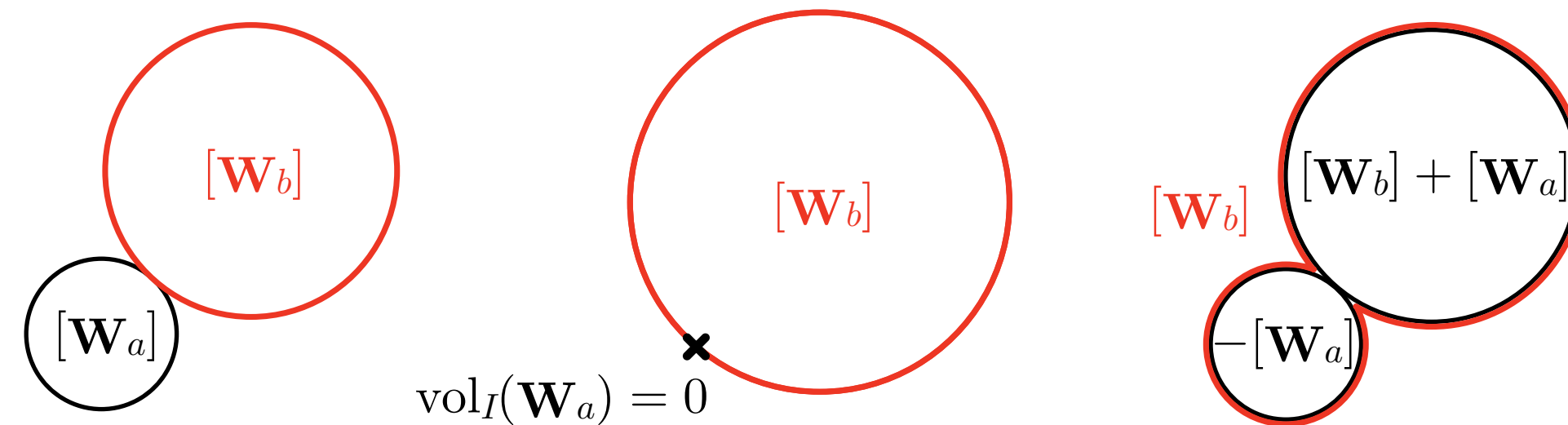
Afine D_4 Dynkin diagram



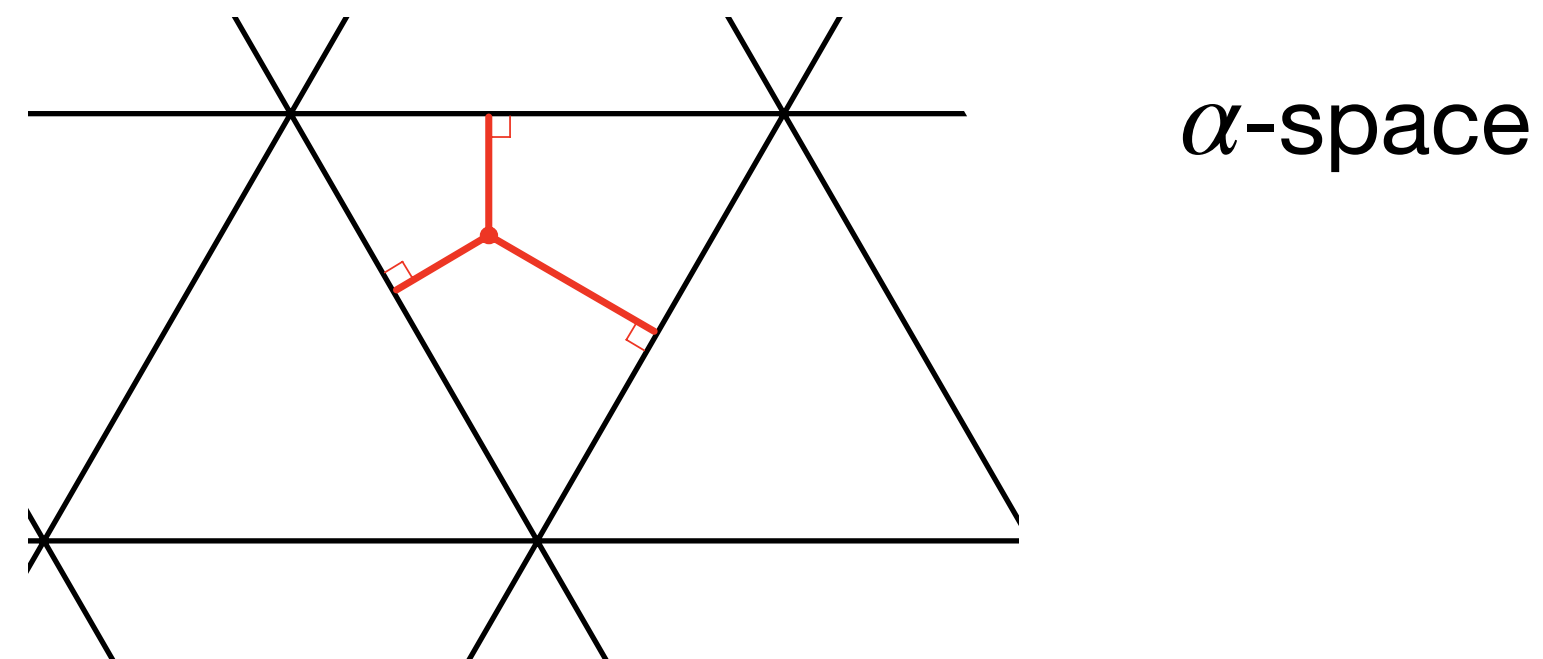
Affine Weyl Group Action via Wall-Crossing

- Affine Weyl group action via Picard-Lefschetz transformation

$$\dot{W}(D_4) \curvearrowright H_2(\mathfrak{X}, \mathbb{Z}) \cong \dot{Q}(D_4) .$$



- Given a chamber, volumes of $\mathbb{P}^1$ are distances to the wall



- At an appropriate chamber, the volumes are given by

$$\text{vol}(\mathbf{D}_i) = \int_{\mathbf{D}_i} \frac{\omega_I}{2\pi} = \alpha \cdot e^i .$$

Compact A-branes and finite-dimensional modules

Dimension formula

$$\ell_i = \dim \mathcal{D}_i = \int_{\mathbf{D}_i} \frac{\Omega_I}{2\pi\hbar} = \frac{\alpha \cdot e^i}{\hbar} \, .$$

Claim The functor $\mathrm{Hom}(-, \mathfrak{B}_{\mathrm{cc}})$ restricts to a derived equivalence of the full subcategory of compact Lagrangian A -branes of $\mathfrak{X}$ and the category of finite-dimensional $S\ddot{H}_{q,t}$ -modules.

finite-dim rep	shortening condition	A -brane	A -brane condition
$\mathcal{F}_m^{(x_m)}$	$q^m = 1$	$\mathfrak{B}_{\mathbf{F}}^{(x_m)}$	$m = \frac{1}{\hbar}$
$\mathcal{D}_{\ell_0}^{(0)}$	$q^{\ell_0} = \boldsymbol{t}^{-e^0} = t_1 t_2 t_3 t_4$	$\mathfrak{B}_{\mathbf{D}_0}$	$\ell_0 = \frac{\alpha \cdot e^0}{\hbar} = \frac{1}{\hbar} - \frac{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_1}^{(1)}$	$q^{\ell_1} = \boldsymbol{t}^{-e^1} = t_1 t_2 t_3^{-1} t_4^{-1}$	$\mathfrak{B}_{\mathbf{D}_1}$	$\ell_1 = \frac{\alpha \cdot e^1}{\hbar} = \frac{-\alpha_1 - \alpha_2 + \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_2}^{(2)}$	$q^{\ell_2} = \boldsymbol{t}^{-e^2} = t_1 t_2^{-1} t_3 t_4^{-1}$	$\mathfrak{B}_{\mathbf{D}_2}$	$\ell_2 = \frac{\alpha \cdot e^2}{\hbar} = \frac{-\alpha_1 + \alpha_2 - \alpha_3 + \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_3}^{(3)}$	$q^{\ell_3} = \boldsymbol{t}^{-e^3} = t_1 t_2^{-1} t_3^{-1} t_4$	$\mathfrak{B}_{\mathbf{D}_3}$	$\ell_3 = \frac{\alpha \cdot e^3}{\hbar} = \frac{-\alpha_1 + \alpha_2 + \alpha_3 - \alpha_4}{\hbar}$
$\mathcal{D}_{\ell_4}^{(4)}$	$q^{\ell_4} = \boldsymbol{t}^{-e^4} = t_1^{-2}$	$\mathfrak{B}_{\mathbf{D}_4}$	$\ell_4 = \frac{\alpha \cdot e^4}{\hbar} = \frac{2\alpha_1}{\hbar}$

Affine Braid Group Action

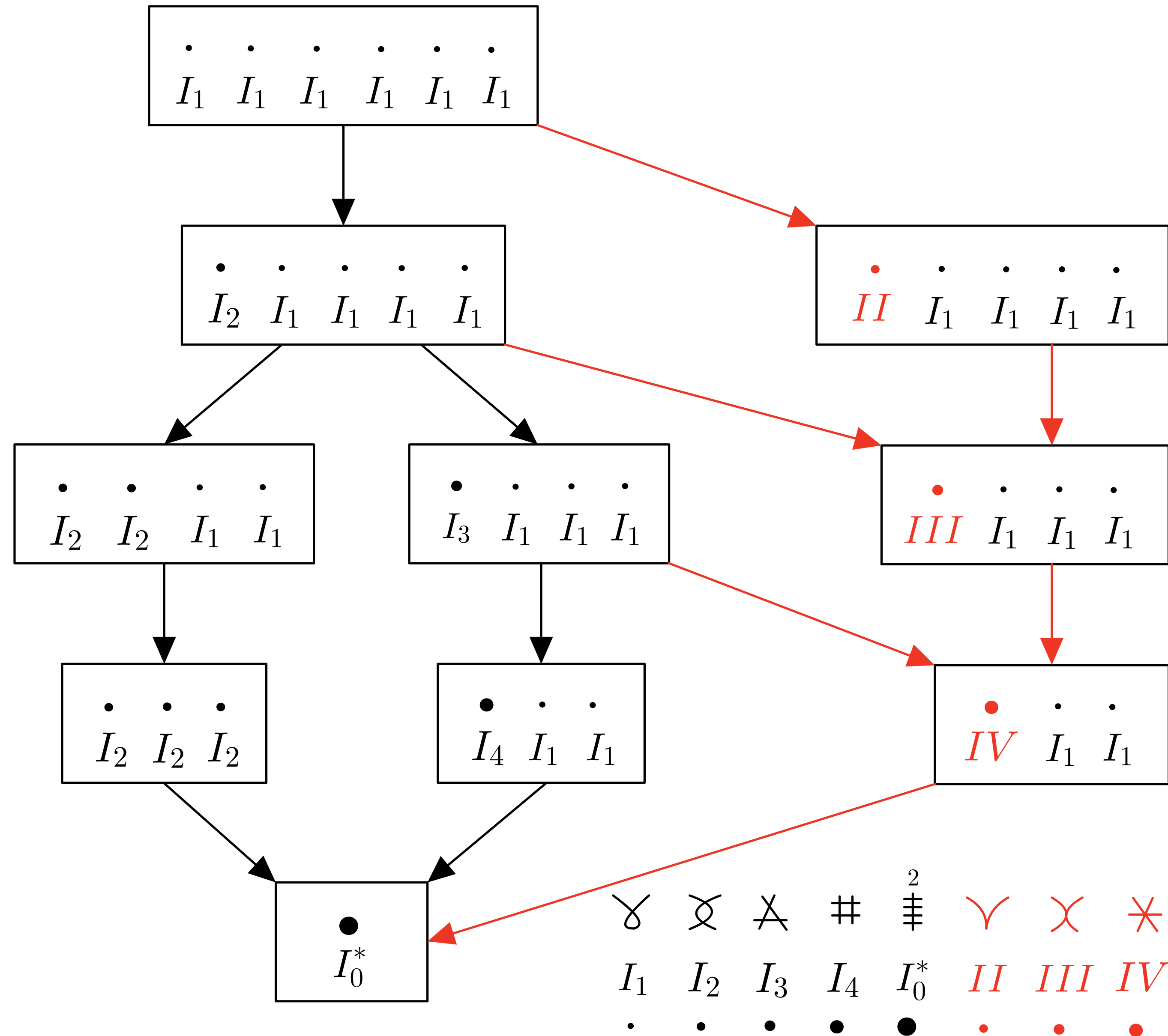
- The A -model depends only on complexified Kähler moduli, but is independent of complex structure.
- The choice of symplectic form $\omega_{\mathfrak{X}}$ depends on $\hbar$.
- If $\hbar \in \mathbb{R}$, then (α, β) is the complex structure, while $\gamma + i\eta$ is the complexified Kähler moduli.
- Variations along a loop in the complex structure moduli space (avoiding singularities) induce non-trivial transformations of the A -brane category.
- When $\gamma + i\eta = 0$, the affine braid group $\dot{\text{Br}}(D_4)$ acts on the category:

$$\begin{array}{ccc} & \dot{\text{Br}}(D_4) & \\ \circlearrowleft & & \circlearrowright \\ A\text{-Brane}(\mathfrak{X}, \omega_{\mathfrak{X}})_{\gamma=0=\eta} & \cong & \text{Rep}(S\ddot{H}_{q,t})_{|q|=1=|t_j|} \end{array}$$

where $T_a([\mathbf{D}_b]) = [\mathbf{D}_b] + ([\mathbf{D}_a] \cdot [\mathbf{D}_b])[\mathbf{D}_a]$ are subject to

$$\begin{aligned} T_j T_k &= T_k T_j \\ T_j T_4 T_j &= T_4 T_j T_4 \end{aligned}$$

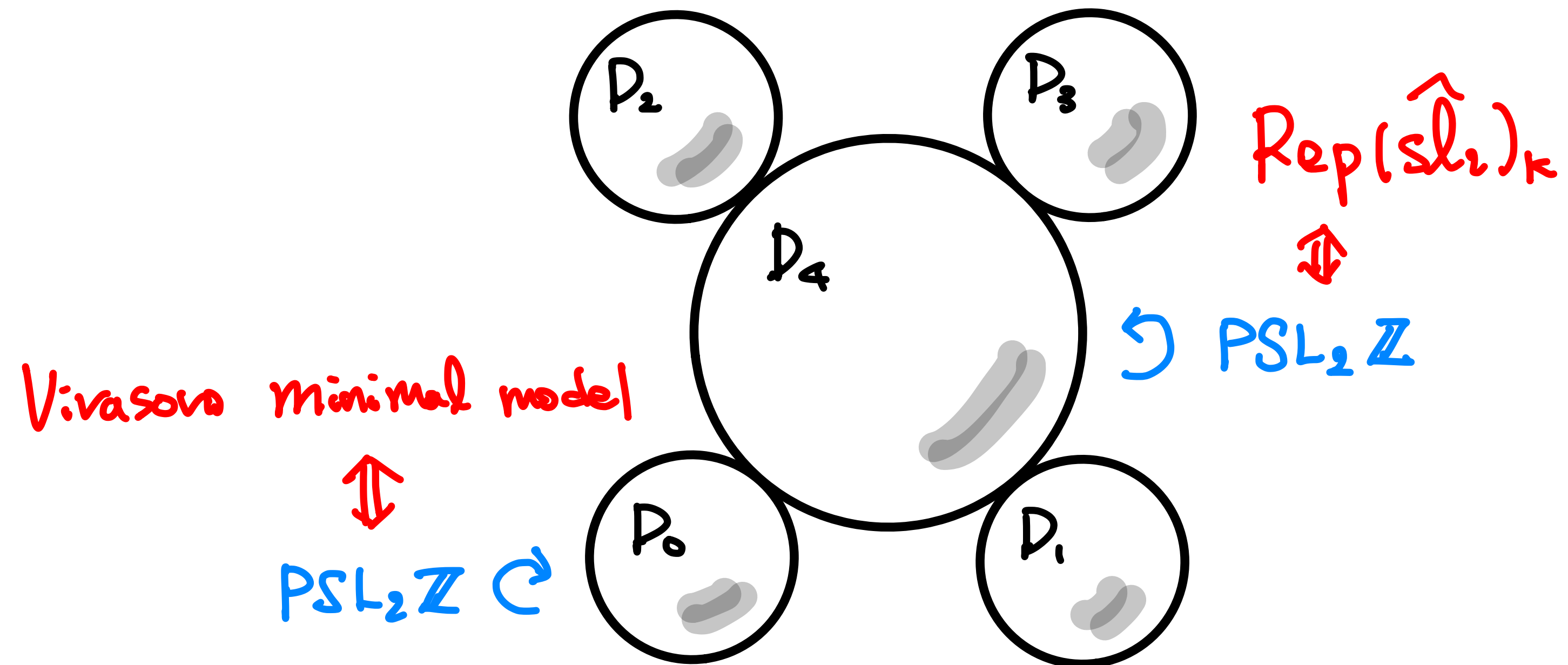
Classification of Kodaira singular fibers



A type limit and modularity

Gukov-Koroteev-SN-Pei-Saberi

$\mathrm{PSL}(2, \mathbb{Z})$ -actions on $\mathfrak{B}_{D_0}$ and $\mathfrak{B}_{D_4}$ are categorified by modular tensor categories



Higher rank generalization

$$K^0(\mathrm{MTC}[W_N(N, M + N)]) \cong \mathrm{Hom}(\mathfrak{B}_{\mathrm{cc}}, \mathfrak{B}_{D_1^{(N)}}) \cong \mathcal{H}_{\mathrm{bottom}}(T_{N, M}) .$$

More generalizations and $\mathbb{C}^\times$ -fixed affine Springer fibers

Shan-Yan-Xie

Future directions

- B-model approach via SYZ mirror symmetry and geometric Langlands duality

Kapustin-Witten, Gukov-Witten

- Quantizations of other character varieties and Coulomb branches

Skein algebras of punctured Riemann surface

Deligne-Cvitanovich exceptional series

Etingof-Oblomkov-Rains

- Higher-rank generalizations

Geometry of Hitchin moduli spaces from DAHA modules?

- BPS quivers and cluster varieties

- Relation to Springer fibers and 4d mirror symmetry?